Homework Assignment Week 9

Objective:

- To be able to read data from a text file
- To be able to store the data in a list
- To be able to use the list methods for calculations
- To be able to handle the exceptions in the code

Instructions:

1. Name your python file as HW_W9_{yourFirstName}{LastNameInitial}.py
   For example: Michael K will name the python file of homework assignment Week 9 as **HW_W9_MichaelK.py**
2. Write the Python code for **either Problem 1 or Problem 2**
3. The code should only use **Python commands that have been covered in the course up to this point.**
4. You will record a small video (3-5 min) to explain the code you wrote for the assignment. For learning how to create a recording in Zoom, refer to document '**How to record video presentation using Zoom**' in **D2L Content -> How-To Guides**
5. Submit **both, the .py file and the video**, in D2L Assignments -> **Homework Assignment Week 9** Dropbox. Refer to the document '**How to submit homework assignment with video presentation'** in **D2L Content -> How-To Guides.**

Grading Rubric:

- The grading rubric is as shown below. You can also find this rubric in D2L under Homework Assignment Week 2.
- 15 points are assigned for the Python file submitted
- 10 points are assigned for the video presentation
- Refer to the document '**Coding Assignment Checklist**' in D2L Content -> How-To Guides to make sure you meet all the grading requirements

Problem 1: Population Data

Homework Assignment Week 9 has a text file named `USPopulation.txt` (file provided in D2L along with homework assignment). This file contains the midyear population of the United States, in thousands, during the years 1950 through 1990. The first line in the file contains the population for 1950, the second line contains the population for 1951 and so forth.

Write a program that reads the file contents into a list. The program should display the following data:

- The average annual change in population during the time period
- The year with the greatest increase in population during the time period
- The year with the smallest increase in population during the time period

Program should be written to handle following exceptions:

- It should handle any `IOError` exceptions that are raised when the file is opened and data is read from it
- It should handle any `ValueError` exceptions that are raised when the items are read from the file and are converted to a number. In case of `ValueError`, that value must be skipped and code should read another data from the file.
- It should handle any `IndexError` exceptions that are raised when code tries to access a non-existent index of list

Expected Output:

Test Case 1: When code runs without exceptions

```
US population from 1950 to 1990= [151868, 153982, 156393, 158956, 161884, 165069, 168088,
171187, 174149, 177135, 179979, 182992, 185771, 188483, 191141, 193526, 195576, 197457,
199399, 201385, 203984, 206827, 209284, 211357, 213342, 215465, 217563, 219760, 222095,
224567, 227225, 229466, 231664, 233792, 235825, 237924, 240133, 242289, 244499, 246819,
249623]

Change in population= [2114, 2411, 2563, 2928, 3185, 3019, 3099, 2962, 2986, 2844, 3013,
2779, 2712, 2658, 2385, 2050, 1881, 1942, 1986, 2599, 2843, 2457, 2073, 1985, 2123, 2098,
2197, 2335, 2472, 2658, 2241, 2198, 2128, 2033, 2099, 2209, 2156, 2210, 2320, 2804]

Average change in population = 2443.875

Year with maximum population increase is: 1955

Year with minimum population increase is: 1967
```

Test Case 2: When `IOError` occurs

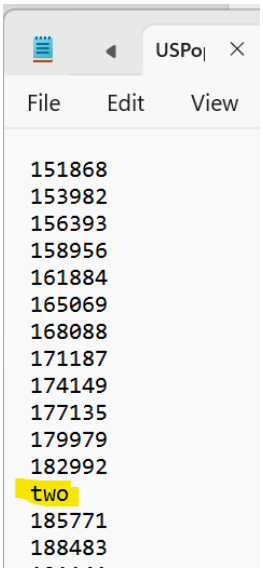
To generate this error, try opening a text file in the code that does not exist, such as USPopulation1.txt.

```
In [1]: runfile('C:/Users/og2360lu/AA Normandale College/AA Intro to Python/AA
Spring 2026/Assessments/Week 10/HW7_Problem1.py', wdir='C:/Users/og2360lu/AA
Normandale College/AA Intro to Python/AA Spring 2026/Assessments/Week 10')
Text file not found.

In [2]:
```

Test Case 3: When `ValueError` occurs

To generate this error, edit the USPopulation.txt file as shown below:



Output:

```
Value is not in correct format. Skipping the value.
US population from 1950 to 1990= [151868, 153982, 156393, 158956, 161884, 165069,
168088, 171187, 174149, 177135, 179979, 182992, 185771, 188483, 191141, 193526,
195576, 197457, 199399, 201385, 203984, 206827, 209284, 211357, 213342, 215465,
217563, 219760, 222095, 224567, 227225, 229466, 231664, 233792, 235825, 237924,
240133, 242289, 244499, 246819, 249623]

Change in population= [2114, 2411, 2563, 2928, 3185, 3019, 3099, 2962, 2986,
2844, 3013, 2779, 2712, 2658, 2385, 2050, 1881, 1942, 1986, 2599, 2843, 2457,
2073, 1985, 2123, 2098, 2197, 2335, 2472, 2658, 2241, 2198, 2128, 2033, 2099,
2209, 2156, 2210, 2320, 2804]

Years= [1951, 1952, 1953, 1954, 1955, 1956, 1957, 1958, 1959, 1960, 1961, 1962,
1963, 1964, 1965, 1966, 1967, 1968, 1969, 1970, 1971, 1972, 1973, 1974, 1975,
1976, 1977, 1978, 1979, 1980, 1981, 1982, 1983, 1984, 1985, 1986, 1987, 1988,
1989, 1990]

Average change in population = 2443.875

Year with maximum population increase is: 1955

Year with minimum population increase is: 1967
```

Test Case 4: When `IndexError` occurs

This error can occur when the code tries to access an element of a list at an index beyond the size of the list. To generate this error, try to print an element at index 90 of the population data list. Since the population list does not have index 90, it should show an index error.

Output:

```
In [4]: runfile('C:/Users/og2360lu/AA Normandale College/AA Intro to Python/AA
Spring 2026/Assessments/Week 10/HW7_Problem1.py', wdir='C:/Users/og2360lu/AA
Normandale College/AA Intro to Python/AA Spring 2026/Assessments/Week 10')
US population from 1950 to 1990= [151868, 153982, 156393, 158956, 161884, 165069,
168088, 171187, 174149, 177135, 179979, 182992, 185771, 188483, 191141, 193526,
195576, 197457, 199399, 201385, 203984, 206827, 209284, 211357, 213342, 215465,
217563, 219760, 222095, 224567, 227225, 229466, 231664, 233792, 235825, 237924,
240133, 242289, 244499, 246819, 249623]
There was an indexing error.
```

Problem 2: 1994 Weekly Gas Graph

Homework Assignment Week 9 has a text file named `1994_Weekly_Gas_Averages.txt`. This file contains the average gas price for each week in the year 1994. (There are 52 lines in the file.) Using `matplotlib`, write a python program that reads the contents of the file then plots the data as

either line graph or bar chart. Be sure to display meaningful labels along the X and Y axes, as well as the tick marks. Hint: Use rotation argument as plt.xticks(week_list,rotation=90) to change the orientation of xtick labels.

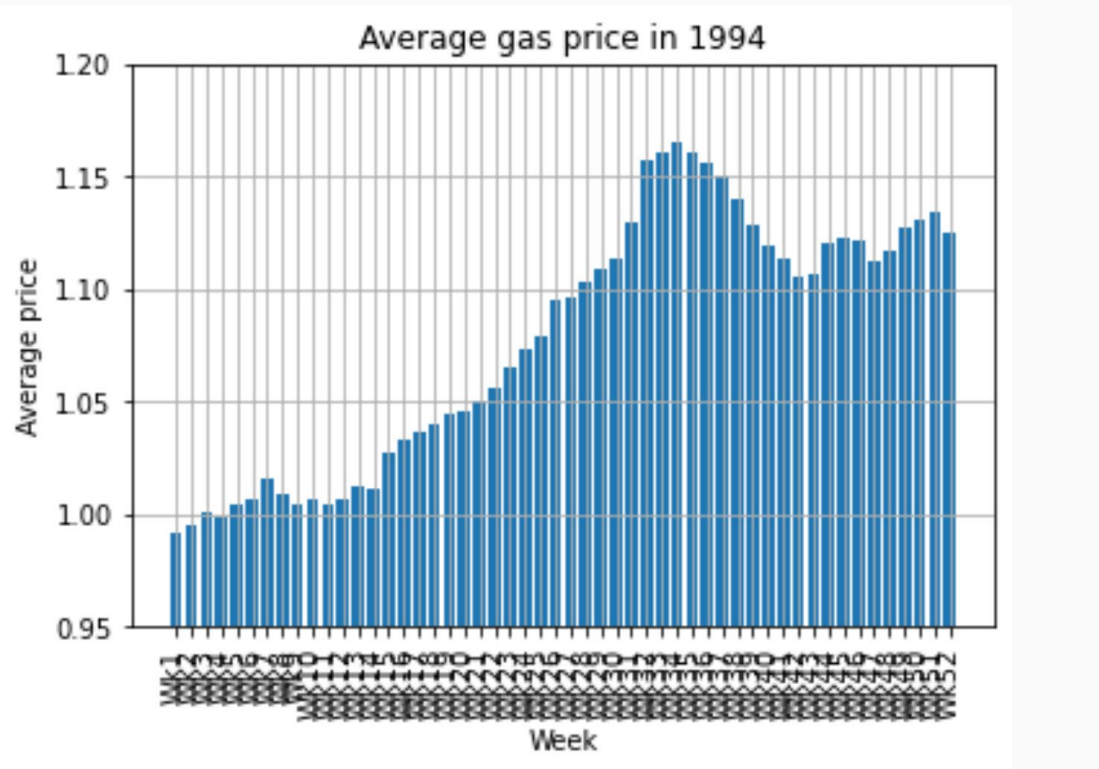
Program should be written to handle following exceptions:

- It should handle any `IOError` exceptions that are raised when the file is opened and data is read from it
- It should handle any `ValueError` exceptions that are raised when the items are read from the file and are converted to a number.
- It should handle any `IndexError` exceptions that are raised when code tries to access a non-existent index of list

Expected Output:

Test Case 1: Code runs without any exceptions

```
Average gas prices= [0.992, 0.995, 1.001, 0.999, 1.005, 1.007, 1.016, 1.009, 1.004, 1.007,
1.005, 1.007, 1.012, 1.011, 1.028, 1.033, 1.037, 1.04, 1.045, 1.046, 1.05, 1.056, 1.065,
1.073, 1.079, 1.095, 1.097, 1.103, 1.109, 1.114, 1.13, 1.157, 1.161, 1.165, 1.161, 1.156,
1.15, 1.14, 1.129, 1.12, 1.114, 1.106, 1.107, 1.121, 1.123, 1.122, 1.113, 1.117, 1.127,
1.131, 1.134, 1.125]
['Wk1', 'Wk2', 'Wk3', 'Wk4', 'Wk5', 'Wk6', 'Wk7', 'Wk8', 'Wk9', 'Wk10', 'Wk11', 'Wk12',
'Wk13', 'Wk14', 'Wk15', 'Wk16', 'Wk17', 'Wk18', 'Wk19', 'Wk20', 'Wk21', 'Wk22', 'Wk23',
'Wk24', 'Wk25', 'Wk26', 'Wk27', 'Wk28', 'Wk29', 'Wk30', 'Wk31', 'Wk32', 'Wk33', 'Wk34',
'Wk35', 'Wk36', 'Wk37', 'Wk38', 'Wk39', 'Wk40', 'Wk41', 'Wk42', 'Wk43', 'Wk44', 'Wk45',
'Wk46', 'Wk47', 'Wk48', 'Wk49', 'Wk50', 'Wk51', 'Wk52']
```



Test Case 2: When `IOError` occurs

To generate this error, try opening a text file in the code that does not exist, such as `1994_Weekly_Gas_Averages1.txt`.

```
In [6]: runfile('C:/Users/og2360lu/AA Normandale College/AA Intro to Python/AA
Spring 2026/Assessments/Week 10/HW7_problem2.py', wdir='C:/Users/og2360lu/AA
Normandale College/AA Intro to Python/AA Spring 2026/Assessments/Week 10')
Text file not found.
```

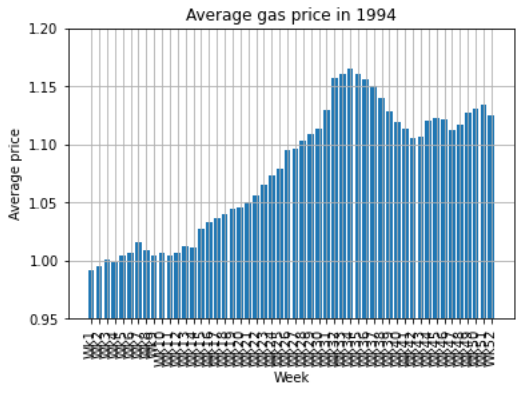
Test Case 3: When `ValueError` occurs

To generate this error, edit the `1994_Weekly_Gas_Averages.txt` file as shown below:

```
0.992
0.995
1.001
0.999
1.005
1.007
1.016
1.009
1.004
1.007
1.005
1.007
1.012
1.011
one
1.028
```

Output:

```
Value is not in correct format. Skipping the value.
Average gas prices= [0.992, 0.995, 1.001, 0.999, 1.005, 1.007, 1.016, 1.009,
1.004, 1.007, 1.005, 1.007, 1.012, 1.011, 1.028, 1.033, 1.037, 1.04, 1.045,
1.046, 1.05, 1.056, 1.065, 1.073, 1.079, 1.095, 1.097, 1.103, 1.109, 1.114, 1.13,
1.157, 1.161, 1.165, 1.161, 1.156, 1.15, 1.14, 1.129, 1.12, 1.114, 1.106, 1.107,
1.121, 1.123, 1.122, 1.113, 1.117, 1.127, 1.131, 1.134, 1.125]
['Wk1', 'Wk2', 'Wk3', 'Wk4', 'Wk5', 'Wk6', 'Wk7', 'Wk8', 'Wk9', 'Wk10', 'Wk11',
'Wk12', 'Wk13', 'Wk14', 'Wk15', 'Wk16', 'Wk17', 'Wk18', 'Wk19', 'Wk20', 'Wk21',
'Wk22', 'Wk23', 'Wk24', 'Wk25', 'Wk26', 'Wk27', 'Wk28', 'Wk29', 'Wk30', 'Wk31',
'Wk32', 'Wk33', 'Wk34', 'Wk35', 'Wk36', 'Wk37', 'Wk38', 'Wk39', 'Wk40', 'Wk41',
'Wk42', 'Wk43', 'Wk44', 'Wk45', 'Wk46', 'Wk47', 'Wk48', 'Wk49', 'Wk50', 'Wk51',
'Wk52']
```



Test Case 4: When `IndexError` occurs

This error can occur when the code tries to access an element of a list at an index beyond the size of the list. To generate this error, try to print an element at index

90 of the gas prices list. Since the gas prices list does not have index 90, it should show an index error.

```
Average gas prices= [0.992, 0.995, 1.001, 0.999, 1.005,
1.007, 1.016, 1.009, 1.004, 1.007, 1.005, 1.007, 1.012,
1.011, 1.028, 1.033, 1.037, 1.04, 1.045, 1.046, 1.05,
1.056, 1.065, 1.073, 1.079, 1.095, 1.097, 1.103, 1.109,
1.114, 1.13, 1.157, 1.161, 1.165, 1.161, 1.156, 1.15, 1.14,
1.129, 1.12, 1.114, 1.106, 1.107, 1.121, 1.123, 1.122,
1.113, 1.117, 1.127, 1.131, 1.134, 1.125]
Index out of range.
```